



Euclid's *Elements* is one of the most important mathematical texts from the ancient world. It consists of 13 books and was originally written in Greek.



“IF YOU CUT OPEN THE **HEAD**, YOU WILL FIND THE **BRAIN** HUMID, FULL OF SWEAT AND HAVING A BAD SMELL...”

Hippocrates, from *On the Sacred Disease*, 400 BCE

Healing hands
A marble frieze showing Hippocrates treating a sick woman. He advocated careful examination to determine the underlying disease.

ASTRONOMERS IN GREECE

were interested in predicting the location of celestial bodies. This led the Greek astronomer Eudoxus of Cnidus (c.408–355 BCE) to develop a geometrical **model of the heavens**, in which the Sun, Moon, and planets moved in a series of 27 concentric spheres. He also made an accurate estimate of the length of the year at 365.25 days. At the time, most Greek astronomers believed Earth was stationary at the center of the Solar System, but Heraclides of Pontus (388–312 BCE) offered a variation

on this theory. He claimed that **Earth rotated on an axis**, which explained the changing seasons.

Greek medicine moved in a more scientific direction when Alcmaeon of Croton began to teach that health is achieved by balancing the elements in the body. Hippocrates of Cos (460–370 BCE), who valued clinical observation, including taking a patient's pulse, applied this theory, teaching that **imbalances in the body** and impurities in the air could cause disease. In the mid-5th century BCE, Euryphon of Cnidus, who was

from a rival school, taught that diseases were caused by residues building up in the body and advised that these be neutralized.

The Greek polymath, Aristotle, **refined the theory of the four elements**—earth, air, fire,

and water—to include a fifth—*aither*—which caused the stars and planets to move in a circular motion. Aristotle modified Eudoxus's theory to explain anomalies, adding additional spheres to a total of 55. He also

MOTION OF THE SPHERES

Greek astronomers explained irregularities in planetary motions by theorizing that the Sun, Moon, and planets each sat in a series of concentric spheres. The circular motion (at differing speeds) of each sphere generated the planet's orbits.

In the early 2nd century, the astronomer Ptolemy replaced the spheres with circles in his model of the Solar System.

began the study of dynamics by theorizing that speed could be directly proportional to the weight of the body, the force applied, and the density of the medium in which the body moved.

The **foundations of geometry** were laid in the mid-4th century BCE by the Greek mathematician and father of geometry, **Euclid of Alexandria** (325–265 BCE), in his 13-book work called *Elements*. In it he puts forward a set of five “geometrical postulates” and nine “common notions” (or axioms). From these he deduced a set of theorems, including Pythagoras's theorem, and that the sum of angles in a triangle is always 180 degrees. *Elements* also included pioneering work on number theory, including an algorithm for the greatest common divisor.

c.400 BCE Hippocrates teaches that **imbalances in the body** can cause disease

c.390–350 BCE Plato puts forward the theory of ideal forms

387 BCE Plato founds the Academy school at Athens

c.375 BCE Eudoxus of Cnidus develops his theory of motion of celestial spheres

c.350–300 BCE Boat from Hjortspring in Denmark is the first clinker-built vessel

c.350 BCE Heraclides of Pontus teaches that Earth rotates on its axis

350–322 BCE Aristotle refines the theory of motion of spheres